TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Audio-Centered Multimodal Framework for Industrial System Understanding			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical · Electronics		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Measuring Instruments	Safety Monitoring / Diagnosis Control Instruments	R&D / Engineering Services	Testing / Inspection / Analysis Techniques
Concept of Proposed Technology	Current industrial monitoring systems rely on multiple heterogeneous sensors, such as vibration, current, and vision, but these modalities are often analyzed in isolation. This fragmented approach limits the ability to understand the overall behavior of complex industrial systems. This project proposes an audio-centered multimodal framework that uses sound as the primary modality for interpreting industrial system behavior, while flexibly incorporating complementary information from other sensor types. Rather than focusing on task-specific detection or classification, the proposed technology aims to build a unified representation layer that enables integrated, interpretable, and reusable understanding of industrial systems across different operating conditions. By positioning audio as a central and non-invasive probe, the framework provides a scalable			

	foundation for multimodal system interpretation, supporting a wide range of monitoring, diagnostic, and analysis applications. The proposed approach is conceptually aligned with prior studies on multimodal machine learning, which emphasize the importance of unified representation spaces for integrating heterogeneous sensing modalities [1].
Rationale for Joint R&D Project	The development of a general-purpose multimodal interpretation framework for industrial systems requires validation across diverse sensing setups, machine types, and operational environments. Such diversity cannot be sufficiently covered by a single institution or a single-country testbed. Through international joint R&D, this project enables systematic investigation of how audio-centered representations can be aligned with heterogeneous sensor modalities under real-world industrial conditions. This collaborative approach reduces technical uncertainty, ensures that the proposed framework is not site- or vendor-specific, and strengthens its applicability to a broad range of industrial scenarios. Furthermore, the complementary roles of research-oriented institutes and application-driven SMEs allow the project to balance foundational methodological development with practical relevance, accelerating the translation of research outcomes into deployable solutions.
Key Words	1. Industrial Acoustics 2. Multimodal Intelligence 3. Industrial AI Frameworks 4. Interpretable AI

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 2-3	End of the Project	Level 5-6
R&D Objectives	The objective of this project is to develop an audio-centered multimodal framework that enables integrated and interpretable understanding of industrial systems by using sound as the primary sensing modality while incorporating complementary information from other sensor types. Specifically, the project aims to *Objective 1.* Design an audio-anchored representation layer that serves as a common reference space for integrating heterogeneous sensor modalities, such as vibration, electrical signals, and other operational data. *Objective 2.* Develop cross-modal alignment and interpretation mechanisms that enable coherent understanding of system behavior across different sensing channels, rather than isolated analysis of individual signals. *Objective 3.* Establish a general and reusable software framework that supports flexible integration of the proposed representations into various industrial monitoring, diagnostic, and analysis scenarios. Related studies on multimodal sensor fusion in industrial contexts have demonstrated the feasibility of integrating audio with other sensing modalities, while also highlighting the limitations of task-specific and tightly coupled solutions [2–4].			
R&D Target Specifications	The target performance specifications of this project are as follows: 1. Cross-Modal Consistency: Achieve stable and consistent system-level representations across heterogeneous sensor modalities, with at least 40% reduction in modality-dependent variation compared to independent single-modal analysis. 2. Robustness Across Operating Conditions: Maintain stable interpretation performance under varying operating and environmental conditions, limiting performance degradation to within 10–15%. 3. Interpretability for Industrial Use: Provide human-interpretable multimodal indicators of system behavior, with at least 80% agreement between automated outputs and expert assessments. 4. Scenario Reusability: Demonstrate applicability to at least three different industrial monitoring scenarios without significant task-specific reconfiguration.			
R&D Plan	**WP1. Audio-Centered Multimodal Representation Design** This WP develops an audio-anchored representation layer that serves as a common reference space for integrating heterogeneous sensor modalities. The focus is on designing a unified structure that allows different sensor signals to be interpreted coherently at the system level. **WP2. Cross-Modal Alignment and System-Level Interpretation** This WP investigates how information from multiple sensing channels can be aligned and jointly interpreted through the proposed audio-centered representation. The goal is to enable consistent understanding of system behavior rather than isolated signal analysis. **WP3. Modular Software Framework Development** This WP implements the developed concepts as a modular and reusable software framework. The framework will be designed for flexible integration into existing industrial monitoring solutions and adaptable to different sensing configurations.			

	WP4. Cross-Scenario Validation and Benchmarking This WP validates the proposed framework across diverse industrial environments and sensor setups. Quantitative evaluation protocols will be established to assess robustness, interpretability, and reusability.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute (Fraunhofer IDMT) will be responsible for leading the foundational methodological development of the proposed audio-centered multimodal framework. This includes designing the core representation structures, developing cross-modal alignment principles, and establishing interpretation mechanisms that enable coherent system-level understanding. In addition, the institute will define evaluation protocols and benchmarking methodologies to assess robustness, interpretability, and reusability across heterogeneous sensing conditions. By focusing on generalizable concepts rather than application-specific solutions, the institute will ensure that the developed framework remains broadly applicable across different industrial domains. The institute will also coordinate joint research activities, facilitate knowledge transfer among consortium partners, and provide scientific guidance to ensure consistency between theoretical development and practical validation.
Expected R&R of Korean SME Partner	The SME partners will be responsible for providing practical industrial perspectives and real-world constraints to guide the development of the proposed audio-centered multimodal framework. This includes supplying representative multimodal datasets, defining realistic operational requirements, and identifying relevant integration points within existing monitoring and diagnostic systems. In addition, the SMEs will participate in the validation of the developed framework under real or near-real industrial conditions. By integrating the proposed methods into their existing solution pipelines, they will assess usability, scalability, and deployment feasibility. Furthermore, the SME partners will contribute to the exploration of commercialization pathways, identifying potential productization strategies and market opportunities. Their involvement ensures that the research outcomes are aligned with industrial needs and can be translated into practical and sustainable solutions.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to deliver a foundational audio-centered multimodal framework that enables integrated and interpretable understanding of industrial systems. The main outcome will be a general and reusable representation layer that allows heterogeneous sensor data to be coherently aligned and analyzed at the system level, rather than in isolation. From a technological perspective, the project will establish new principles for multimodal system interpretation, positioning sound as a central and non-invasive sensing modality. This will enable more flexible, explainable, and scalable industrial monitoring solutions that are not tied to specific machines or tasks. From an industrial and economic perspective, the proposed framework is expected to reduce system integration complexity, shorten development cycles for new applications, and improve the reliability of data-driven decision-making. By providing a common interpretation layer across modalities, the project will contribute to more sustainable and cost-effective deployment of industrial AI solutions.

6. References

(1) Baltrušaitis, Tadas, Chaitanya Ahuja, and Louis-Philippe Morency. "Multimodal machine learning: A survey and taxonomy." IEEE transactions on pattern analysis and machine intelligence 41.2 (2018): 423-443.

(2) Yan, Jin, et al. "Fusion of audio and vibration signals for bearing fault diagnosis based on a quadratic convolution neural network." Sensors 23.22 (2023): 9155.

(3) Kullu, Omer, and Eyup Cinar. "A deep-learning-based multi-modal sensor fusion approach for detection of equipment faults." Machines 10.11 (2022): 1105.

(4) Rahmati, Milad, and Nima Rahmati. "A multimodal deep learning framework for real-time defect recognition in industrial components using visual, acoustic and vibration signals." Journal of Intelligent Manufacturing and Special Equipment 6.3 (2025): 273-292.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization